8 July 1974
SUPERSEDING
MIL-A-25719(USAF)
14 February 1957

MILITARY SPECIFICATION

ACCELEROMETER, AIRCRAFT, PILOT'S WARNING, TYPE MA-1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of 1-7/8 inch dial, -2 to +4 g range accelerometer, designated Type MA-1.

2. APPLICABLE DOCUMENTS

*2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-636	Boxes, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-C-5541	Chemical Conversion Coatings for Aluminum and Aluminum Alloys
MIL-S-6872	Soldering Process, General Specification For
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-L-25142	Luminescent Material, Fluorescent

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking For Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property

FSC: 6610

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MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection Of
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing Of
MIL-STD-838	Lubrication of Military Equipment
MIL-STD-889	Dissimilar Metals
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form Of
MS33585	Pointers, Dial, Standard Design of Aircraft Instrument
MS33638	Cases, Instrument, Flange Mounted, Aircraft

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

*3.1 Qualification. Accelerometers furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.5 and 6.4).

*3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and service not specified herein shall be selected in accordance with MIL-STD-143.

3.2.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.2.2 Standard parts. With the exception in 3.2, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

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3.3 Materials.

3.3.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.3.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts except where magnetic materials are essential.

3.3.3 Nonferrous materials. Nonferrous materials shall be used for all parts of the accelerometer except where ferrous materials are essential.

3.3.4 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

*3.3.5 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.6 Corrosive fumes. The materials as installed in the accelerometer shall not liberate deleterious fumes.

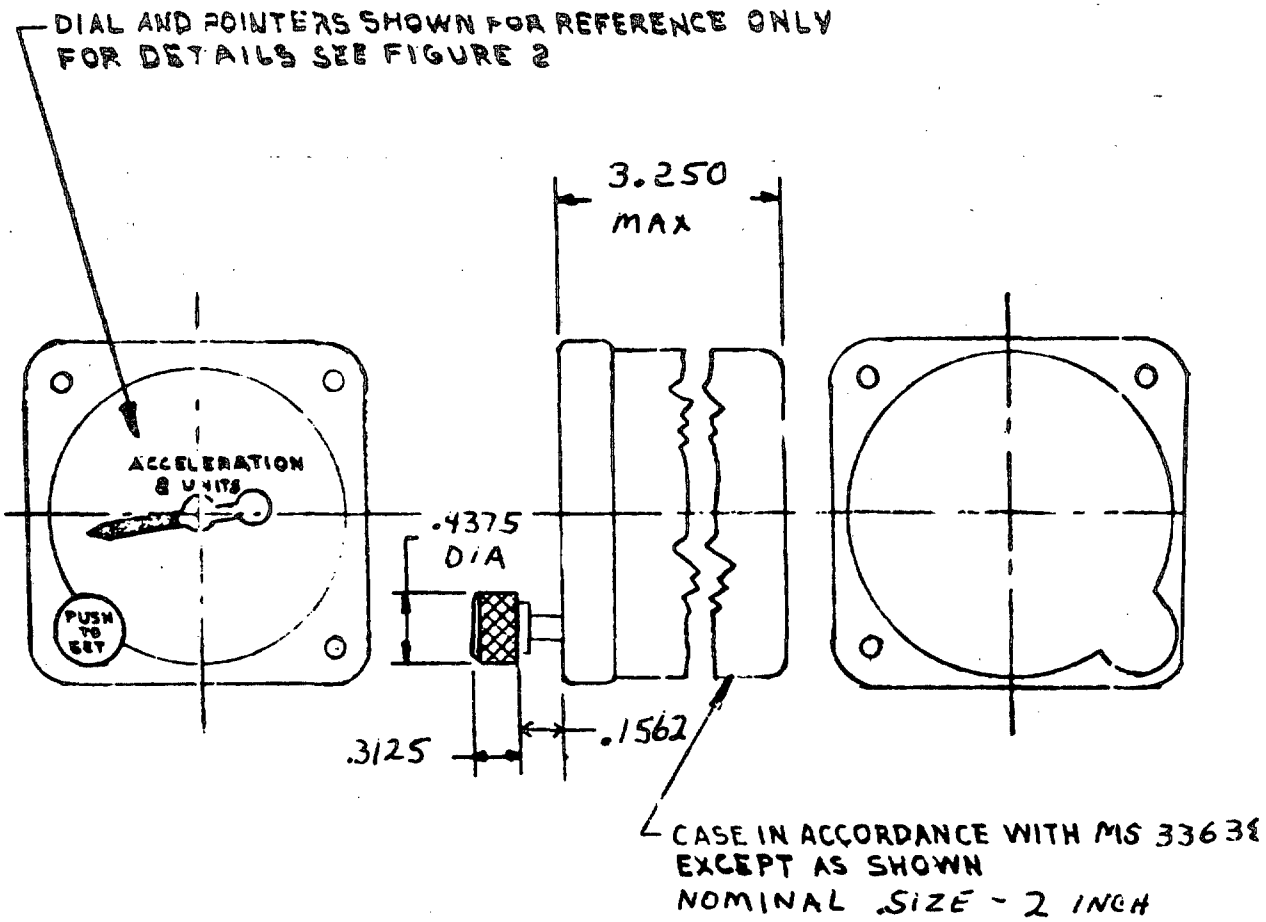
3.3.7 Protective treatment. When materials are used in the construction of the accelerometer that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction.

3.4.1 The accelerometer shall be designed to indicate aircraft accelerations in maneuvers and rough air and shall conform to figure 1.

3.4.1.1 The accelerometer shall be designed with a natural frequency of 6 to 8 cps (cycles per second) and 0.5 to 0.7 critical damping.

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DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED
TOLERANCES ± .016

FIGURE 1 CASE

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3.4.2 The construction of the accelerometer shall be reasonably simple to permit overhaul or repair without the use of special tools or fixtures.

3.4.3 Indication and range. The accelerometer shall indicate vertical accelerations when the plane of the mounting lugs is vertical and the +1 graduation is at 270°. When in this position, normal gravity shall be indicated at +1 g. The accelerometer shall be designed to indicate accelerations in the range -2 to +4 g units.

3.5 Performance. The accelerometer shall be capable of meeting the test requirements specified in Section 4 when subjected to the following conditions:

a. Temperature.-Temperature ranging from +71°C to -54°C.

b. Humidity.-Relative humidity up to 100 percent.

c. Altitude.-Pressures ranging from 30 inches Hg down to 1.31 inches Hg approximately 70,000 feet altitude.

d. Fungus.-Fungus growth as encountered in tropical climates.

e. Thermal shock.-Thermal shock with alternate immersion in water at temperatures of +85°C and +5°C for 8 cycles.

f. Sealing.-Immersion in water with pressure applied and without leakage.

g. Acceleration and shock.-Acceleration and shock forces as anticipated.

h. Vibration error.-When vibrated at an amplitude of 0.010 to 0.020 inch and at a frequency variation between 500 and 3,000 cpm for a total of 3 hours the main pointer shall not exceed the 0.2 g to 0.4 g range and the auxiliary pointers shall not change more than 0.2 g.

i. Vibration endurance.-The dynamic scale error shall not exceed 0.2 g when subjected to the vibration endurance test.

j. Static scale error and friction.-Static scale error and friction shall not exceed $\pm$0.1 g and $\pm$0.2 g, respectively when tested as specified.

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k. Dynamic scale error.-Dynamic scale error shall not exceed ±0.2 g at any specified test point and ±0.1 g at 2/3 of the test points.

l. Magnetic effect.-With the accelerometer revolved about a magnetic compass with a magnetic field intensity of 0.17 to 0.19 oersted, the compass shall not deflect more than 1°.

m. Mounting lugs.-The mounting lugs shall withstand a load of 175 pounds without fracture.

n. Response.-Response characteristics shall be within the limits specified in 4.4.5, when the accelerometer is vibrated at frequencies from 1 to 20 cps at maximum g forces for each test point.

3.5.1 The accelerometer shall be so constructed as to withstand the normal strains, jars, vibrations, and such other conditions as are incident to shipping, storage, installation, and service, without failure.

3.5.1.1 Pivots, bearings, and gears shall neither bind nor shake and shall be as near frictionless as practicable.

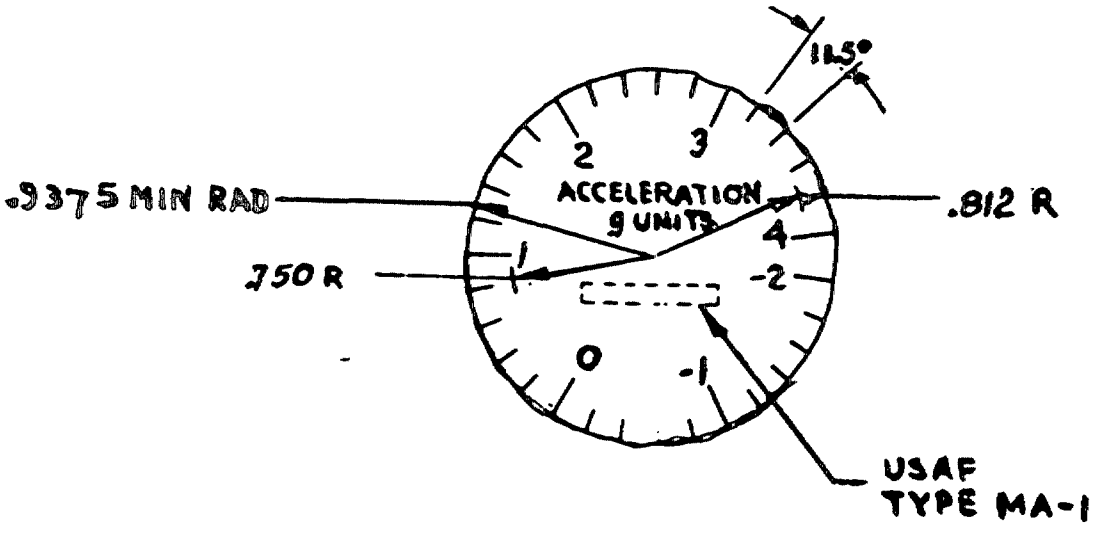
*3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

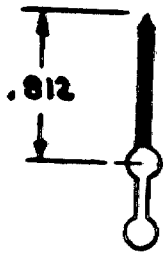
3.7 Dial. The dial shall conform to figure 2 and shall be fastened securely to the case or upon the frame of the mechanism by not less than two screws in such a manner that it will not loosen or turn when the instrument is vibrated.

3.7.1 Distance from dial to cover glass. Distance between dial and cover glass shall not be greater than 0.250 inch.

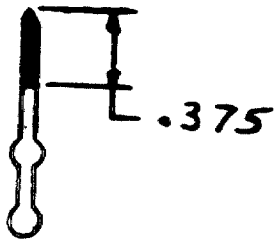
3.7.2 Numerals and letters. The style and proportion of numerals and letters, except the letter "g", placed on the dial of the accelerometer shall conform to MS33558. Numerals shall distinctly indicate the graduation to which each applies.

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MAIN POINTER
MS 33585-3



AUXILIARY POINTER
MS 33585-2
EXCEPT AS SHOWN

DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED
DECIMALS ±0.010
ANGLES ±0° 10'

FIGURE 2. DIAL AND POINTERS

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3.8 Pointers. The accelerometer shall have three pointers. One main pointer for indicating instantaneous acceleration and two auxiliary pointers; one for indicating maximum positive acceleration; and one for indicating maximum negative acceleration. The pointers shall conform to figure 2. They shall be light and sufficiently rigid to prevent oscillation under vibration. They shall be firmly attached to the mechanism, yet shall be readily adjustable.

3.8.1 Ratchet mechanism. Each auxiliary pointer shall be provided with a ratchet device, in order that indication of maximum plus and minus deflections of the main pointer will be maintained until the auxiliary pointers are reset to obtain a new reading.

3.8.2 Auxiliary pointer return mechanism. A mechanism shall be provided for resetting the auxiliary pointers into juxtaposition with the main pointer. A knob, located as shown on figure 1, shall be provided for operating the return mechanism. Stops shall be provided in order that the auxiliary pointers when reset shall not indicate a lower or higher reading than that indicated by the main pointer. Stops shall be a part of the mechanism connecting the pointers and shall not directly engage them. Operation of the return mechanism shall require 2-1/2 to 8 pounds of force on the knob.

3.8.3 Main pointer mechanism. The main pointer mechanism shall not engage the mechanism of the auxiliary pointers when the imposed acceleration is less than the acceleration indicated by the auxiliary pointers.

3.9 Acceleration-Responsive element.

3.9.1 Stops for acceleration-responsive element. Mechanical stops shall be provided for arresting the motion of the acceleration-responsive element when the accelerometer is subjected to accelerations beyond the limits specified. One stop shall engage the element at an acceleration of not less than -2.1 g and not more than -2.2 g, and the other at an acceleration of not less than +4.1 g and not more than +4.2 g.

*3.9.2 Locking device. A device for manually locking the position of the acceleration-responsive element shall be provided, if the design is such as to require it in order to protect the mechanism of the accelerometer from possible damage during shipment. If provided, this device shall be controlled by means of a knurled knob located in the back of the case.

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3.9.3 Adjustment. The mechanism shall be provided with means to adjust or correct the indications. This means of adjustment shall be reasonably simple to manipulate and of such a nature that adjustment can be accomplished with tools ordinarily possessed by an instrument repairman.

3.10 Case. The case shall be in accordance with figure 1 and the requirements specified herein. It shall provide a hermetically sealed inclosure for the mechanism of the instrument. The design of the case shall be such that the mechanism may be removed from the case, replaced and the case resealed. This shall be accomplished without the use of special tools and fixtures unless they are approved by the procuring activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of any atmosphere to which the indicator may be subjected. The case shall be so designed that it will meet the pressure requirements necessary when tested for leaks by means of a mass spectrometer type of helium leak detector as specified in 4.4.4.

3.10.1 Filling medium. In the hermetically sealed case the filling medium shall be of at least 98 percent purity, free of dust particles and containing not more than 0.006 milligrams of water vapor per liter (dew point -65°C) at the filling pressure. The filling medium shall be either 100 percent helium or a mixture of 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be approximately one atmosphere or one-half atmosphere as required. Where practicable the 100 percent helium filling medium shall be utilized.

3.11 Marking. All markings shall be durable to withstand the usage encountered in service. Numerals shall distinctly indicate the graduation to which each applies.

*3.11.1 Fluorescent-Luminescent material. The following markings shall be finished in fluorescent-luminescent material in accordance with MIL-L-25142, type I or III, as applicable. The dimensions of the markings shall be as follows:

	Height or Length Inch ±.015625	Width of Line or Graduation Inch ±0.005
Numerals	.1875	MS33558
Graduations corresponding to numerals	See Figure 2	0.031
Lettering "g UNITS" and "ACCELERATION"	.0625	MS33558
One "g" Graduation	-----	0.062
Shaded part of pointers	-----	-----
Knob markings	-----	-----
Small graduations	See Figure 2	0.015

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*3.11.2 Durable dull black. The following markings shall be durably and legibly marked on the dial. These markings, the background of the dial, and all other markings not otherwise specified shall be finished in durable dull black.

	Height or Length Inch ±.015625
Lettering "USAF" and "TYPE MA-1"	.0625

3.11.3 Visibility of dial. The pointers, numerals, at least 1/16 inch of the shortest graduations, and other specified markings on the dial shall be visible from any point within the frustum of a cone whose side makes an angle of 30° with a perpendicular to the dial and whose small diameter is the aperture of the instrument case.

3.12 Soldering. Soldering shall be accomplished in accordance with MIL-S-6872.

*3.13 Lubrication. The accelerometer shall be lubricated in accordance with MIL-STD-838.

3.14 Weight. The weight of each accelerometer shall not exceed 1.00 pounds.

3.15 Finishes and protective coatings.

3.15.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625 except as follows:

3.15.1.1 Dials, small holes, and case inserts need not be anodized.

3.15.1.2 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.15.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.15.1.4 Castings containing non-aluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.15.1.5 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

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3.15.2 Steel parts. Steel parts shall be cadmium plated where practicable in accordance with QQ-P-416, types II or III, as applicable and of a class that is adequate to achieve the degree of protection required.

3.16 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.17 Workmanship.

3.17.1 General. The accelerometer, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, and riveting; alignment of parts and tightness of assembly screws, and bolts; etc.

3.17.2 Screw assemblies. Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt or threads.

3.17.3 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.17.4 Gears. Gear assemblies shall be properly aligned and meshed, and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustments, gear assemblies shall be free from backlash.

3.17.5 Cleaning. The accelerometer shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material, after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

3.17.6 Dimensions. Dimensions and tolerances not specified, shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the accelerometer they shall be held or limited accordingly.

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3.18 Envelope. An envelope, containing the mounting screws, shall be furnished with each indicator. The following information shall be printed on the face of the envelope:

IMPORTANT
THIS ENVELOPE CONTAINS
MOUNTING SCREWS

4. QUALITY ASSURANCE PROVISIONS

*4.1 Classification of inspections. The inspection and testing of the accelerometer shall be classified as follows:

a. Acceptance inspection.-Acceptance tests are those tests accomplished on accelerometers submitted for acceptance under contract.

b. Qualification inspection.-Qualification tests are those tests accomplished on samples submitted for qualification as a satisfactory product. (See 4.5)

4.2 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests

b. Sampling plans A and B

*4.2.1 Individual tests. Each accelerometer shall be subjected to the following tests as described under 4.4 TEST METHODS:

a. Examination of product

b. Static scale error and friction

c. Dynamic scale error

d. Sealing

e. Response characteristics

f. Return mechanism operation

4.2.2 Sampling plans and tests.

4.2.2.1 Sampling Plan A. One accelerometer selected at random from each 100 or less produced on the contract or order shall be subjected to the following tests as described under 4.4 TEST METHODS:

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a. Individual tests

b. Low temperature

c. High temperature

d. Magnetic effect

e. Vibration error

4.2.2.2 Sampling Plan B. Unless otherwise specified, three accelerometers selected at random from the first 15 items of the contract or order shall be subjected to the following tests as described under 4.4 TEST METHODS:

a. Sampling Plan A

b. Vibration endurance

c. Shock

d. Mounting lugs

e. Low temperature - high altitude

f. Humidity

g. Fungus

h. Thermal shock

i. High temperature exposure

4.2.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined.

4.2.2.3.1 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.2.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

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4.3 Test conditions.

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

*4.3.2 Tapping. Unless otherwise specified, the accelerometer shall be tested in the normal operating position and shall be gently tapped or vibrated before a test reading is taken. Tapping shall be applied in a longitudinal direction with respect to the dial of the instrument. (See 6.2.1)

4.3.3 Vibration stand. A vibration stand shall be used which will vibrate at any desired frequency between 500 and 3000 cpm and shall subject the accelerometer to vibration such that a point on the case will describe in a plane inclined 45° to the horizontal plane, a circle, the diameter of which is equal to the amplitude specified herein.

4.3.4 No adjustments or repairs of any type shall be performed on the qualification test samples after start of qualification tests.

4.4 Test methods.

4.4.1 Examination of product. Each accelerometer shall be examined externally to determine conformance with this specification with respect to material, workmanship, marking, and envelope.

*4.4.2 Static scale error and friction. The position error shall be observed with accelerometer held in each of the following positions:

a. Plane of dial vertical with +1 g graduation at 270°.

b. Plane of dial horizontal.

c. Plane of dial vertical with +1 g graduation at 90°.

In each position, the main-pointer reading obtained shall be due entirely to gravity and not to motion in change of position of instrument. The static scale error shall not exceed ±0.1 g in any of the three postions tested. The accelerometer shall also be tested for friction at the three positions

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mentioned above. The amount of tapping or vibration applied shall be sufficient to remove all friction from the accelerometer. In going from one position to the next, one of the memory pointers shall be engaged. After the desired test position is reached two readings shall be taken, the first before vibration and the second after vibration. The difference of two such readings shall be recorded as friction and shall not exceed 0.2 g with the plane of the dial vertical and 0.3 g with the plane of the dial horizontal.

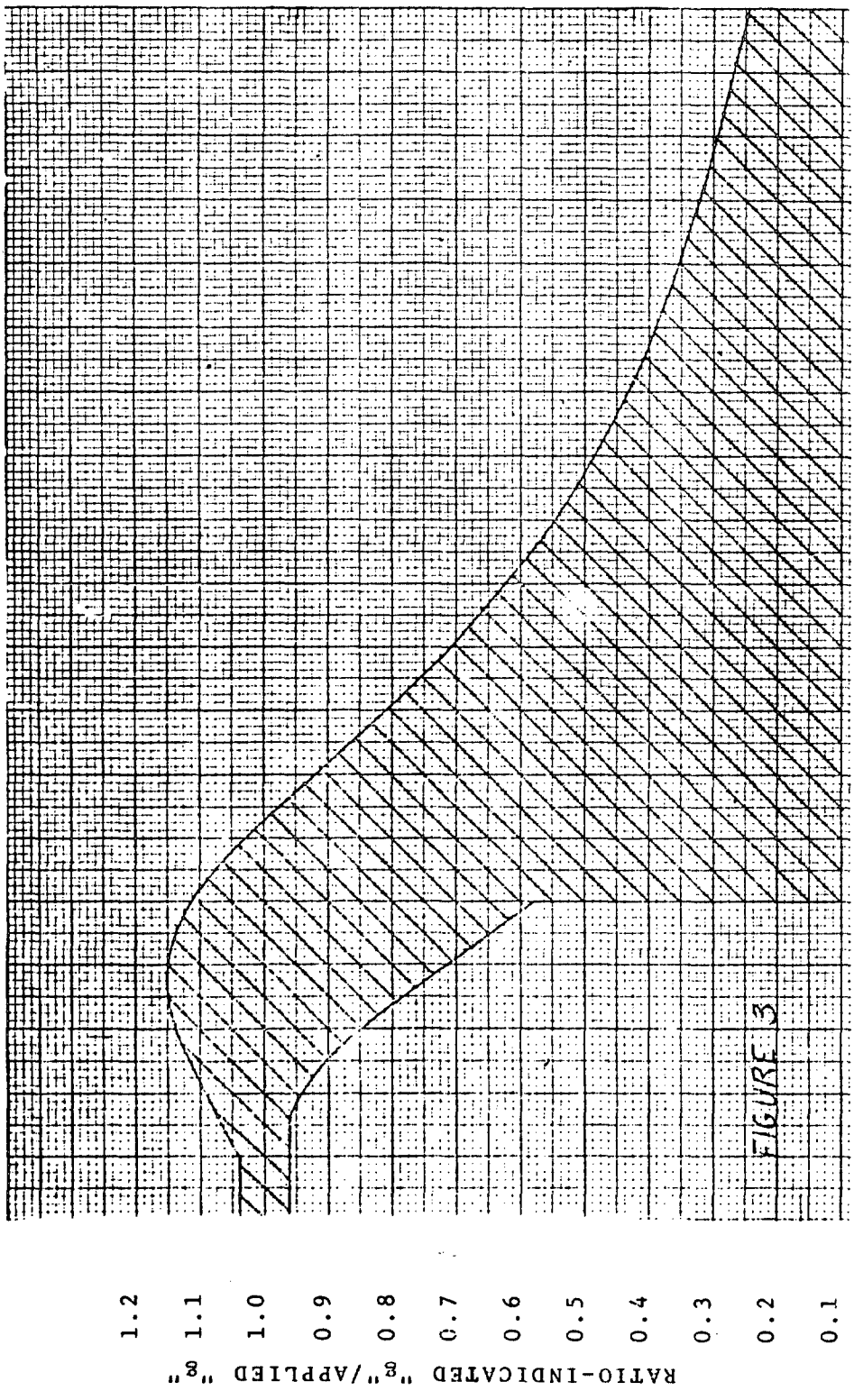
4.4.3 Dynamic scale error. The accelerometer shall be mounted on a suitable centrifuge and the scale errors determined for accelerations of -2, -1, +1, +2, +3, and +4 and at any other points deemed necessary to determine accuracy. The steady state scale error shall not exceed $\pm$0.1 g for 2/3 of the test points.

4.4.4 Sealing. The accelerometer case shall be tested for leaks by means of a mass spectrometer type of helium leak detector and the differential pressure existing when the test is conducted shall be the same as that to which the instrument is filled, approximately one atmosphere. The leak rate shall be based on one atmosphere differential pressure and pure helium as the filling medium. Under these conditions the leak rate shall not exceed 0.10 micron cubic foot per hour. Equivalent helium leak rates for the same hole size at other differential pressures and helium-nitrogen mixtures, or both, shall be calculated by the manufacturer.

4.4.5 Response characteristics. The accelerometer shall be subjected to vibration along an axis parallel to the motion of the sensing element and with the plane of the dial vertical. The test spparatus used shall be as specified in 4.4.5.1. The accelerometer shall be tested at each of a minimum of 16 different frequencies. These tests shall be approximately equally spaced through the frequency range from 4 to 20 cps. The amplitude of vibration at each test point shall be such that the maximum ζ forces practicable for that test point shall be obtained. The indicated g shall be recorded and a plot made for each test point of indicated g/applied g vs frequency. The points so plotted shall fall within the shaded area of graph on figure 3. These graphs shall be a part of the qualification test data.

4.4.5.1 Test apparatus. The equipment shall impose linear sinusoidal motion to the accelerometer and shall be capable of procuring the frequencies and double amplitude required for each test point. The acceleration vector of any harmonic measured on the test table shall not exceed 10 percent of the fundamental acceleration vector.

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4.4.6 Low temperature. The accelerometer shall be subjected to the low temperature test, procedure I of MIL-E-5272. Upon completion of the test period and while at the test temperature the accelerometer shall be subjected to and meet the individual tests.

4.4.7 High temperature. The accelerometer shall be subjected to the high temperature test, procedure I of MIL-E-5272 for a period of 4 hours. Upon completion of the test period and while at the test temperature the accelerometer shall be subject to and meet the individual tests.

4.4.8 Magnetic effect. The accelerometer shall be revolved about a short bar magnet compass with the nearest part of the indicator 5-1/2 inches from the bar magnet, and in a vertical plane which is a perpendicular disector of the North-South axis of the bar magnet. Starting directly under the compass, the indicator shall be held in positions, 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions, the indicator shall be rotated on its horizontal axis until it is in its normal operating position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed 1°.

4.4.9 Vibration error. The accelerometer shall be mounted on a suitable vibration stand with the dial in a vertical plane and with the graduation +1 at 270°. The auxiliary pointers shall be displaced approximately 1 g from the main pointer and the readings noted. The reset mechanism shall not be operated until the completion of the tests. For a total of 3 hours the accelerometer shall be vibrated at the specified frequencies at an amplitude between 0.018 and 0.020 inch, and the pointer oscillation shall be noted. The frequency shall be varied from 500 to 3000 cycles per minute in a period of not more than 60 minutes. The natural frequency of the seismic system shall be noted during the frequency variation between 500 and 3000 cpm. For the remainder of the 3 hour test period, the accelerometer shall be vibrated at a frequency of 1,800 cpm. At the end of the test period, the indication of the auxiliary pointer shall be read before operating the reset mechanism. At frequencies below 1,200 cpm the oscillation of the main pointer shall not exceed 0.2 g and at frequencies over 1,200 cpm, the oscillation shall not exceed 0.3 g. After vibration, the indications of the auxiliary pointers shall not be changed by more than 0.1 g. The accelerometer shall not be damaged as a result of this test.

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4.4.10 Vibration endurance. The accelerometer shall be subjected to the vibration test, procedure V of MIL-E-5272 for a period of 50 hours. Upon completion of the test period the accelerometer shall be subjected to the test specified in 4.4.3. The dynamic scale error shall not exceed 0.2 g.

*4.4.11 Shock. The accelerometer to be installed in the normal operating position is a free-fall apparatus as described in 4.4.11.1. If provided with a locking device, the accelerometer mechanism shall be locked. The accelerometer shall be allowed to fall from a height of 8 inches, then examined for damage and subjected to the test specified in 4.4.3. There shall be no damage to the accelerometer and the steady state scale error shall not exceed 0.2 g.

4.4.11.1 Apparatus. A rigidly constructed device provided with a panel mounting for installing the accelerometer in the specified position, and vertical guides to allow free-fall from the specified height shall be used. A bumper on the free-fall element shall arrest its fall by striking the center of a rounded projection on a metal cross section with an unsupported length of 5 inches between points of attachment to the free-fall element. The bumper shall contact the metal block at the midpoint of a broad side. The center of gravity of the instrument under test and the contact point of the bumper shall be aligned vertically within 1/8 inch. The weight of the free-fall element shall be approximately 12 pounds. The metal block shall weigh not less than 50 pounds.

4.4.12 Mounting lugs. The accelerometer case with mechanism and cover glass removed shall be mounted face downward on the movable head of a suitable testing machine with the diameter of the case in a horizontal plane so that the mounting lugs receive no added support. A suitable pin shall be inserted through the hole in the mounting lug and attached to a pull strap in the stationary head of the machine. A load of 175 pounds shall be applied along the axis of the mounting hole and toward the front of the case for a period of 1 minute. There shall be no fracture.

4.4.13 Low temperature - High altitude. The accelerometer shall be subjected to the low temperature test, procedure II of MIL-E-5272 and an altitude of 70,000 feet (1.31 inches Hg). There shall be no damage to the accelerometer as a result of this test.

4.4.14 Humidity. The accelerometer shall be subjected to the humidity test in accordance with procedure I of MIL-E-5272.

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4.4.15 Fungus. The accelerometer shall be subjected to the fungus resistance test in accordance with procedure I of MIL-E-5272.

4.4.16 Thermal shock cycling. The accelerometer shall be alternately subjected to a total of 8 cycles of immersion in tap water maintained at 85° ±2°C and 5° ±2°C. The length of time for each immersion shall be 30 minutes; not more than 5 seconds shall elapse between immersions. No evidence of moisture penetration or damage to the inclosure shall result from this test. Following this test, the accelerometer shall be subjected to and meet the requirements for the test specified in 4.4.4.

4.4.17 High temperature exposure. The accelerometer shall be subjected to the high temperature test, procedure I of MIL-E-5272. No damage to the accelerometer shall result from this test.

*4.4.18 Return mechanism operation. The accelerometer shall be placed in operating position with the pointers deflected to the maximum indication, the reset knob shall then be operated to return the pointers to zero. The force applied to the reset knob shall be between 2-1/2 and 8 pounds.

4.5 Qualification testing.

4.5.1 Qualification test samples. The qualification test samples shall consist of three accelerometers. Samples shall be identified with the manufacturer's own part number and any additional information required by the letter of authorization.

4.5.2 Qualification required. Prior to actual procurement, the product which this specification covers shall pass the qualification tests specified herein. If the product is later modified in any way, the modified form shall be subjected to and shall pass the same qualification tests.

4.5.3 Qualification tests. The qualification tests shall consist of the tests specified under 4.4 TEST METHODS.

4.6 Inspection and test. Tests for methods of preservation and packaging will be accomplished in accordance with Section 4 of MIL-P-116 to assure compliance with Section 5 of this specification.

5. PREPARATION FOR DELIVERY.-Unless otherwise specified in the individual contract or order the following shall apply:

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*5.1 Preservation and packaging.

*5.1.1 Level A. (Reference definition in MIL-STD-794.)Accelerometer shall be preserved and packaged in quanity unit packs of one each in accordance with Method IC-1 of MIL-P-116. Overbox completed pack in a PPP-B-636 Fiberboard container Weather-Resistant Class. Cushioning material between bag and unit container to be of a type, density and thickness that will insure shock transmission does not exceed the peak values in Gs established for the item when completed packs are subjected to the rough handling drop tests stipulated by MIL-STD-794.

*5.1.2 Level C. (Reference definition in MIL-STD-794.)Accelerometers shall be packaged identically to Level A requirements, except Method III (MIL-P-116) applies in lieu of Method IC-1 and domestic class, PPP-B-636 Box shall be used as the unit container.

5.2 Packing.

5.2.1 Level A. Accelerometers, packaged as specified in 5.1.1, shall be packed in shipping containers conforming to PPP-B-636, Weather-Resistant Class. Insofar as possible, shipping containers shall be uniform shape, size, of minimum tare and cube consistent with the protection required.

5.2.2 Level B. Accelerometers, packaged as specified in 5.1.1, shall be packed in PPP-B-636 Domestic Class boxes. Other requirements specified in 5.2.1 shall apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection against physical and mechanical damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirements of this level.

5.3 Marking.

5.3.1 Container markings. Unit package and shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

"Accelerometer, Aircraft, Pilot's Warning, Type MA-1"

6. NOTES

6.1 Intended use. The Type MA-1 accelerometer covered by this specification is intended for use on aircraft to indicate vertical acceleration in maneuvers and in rough air.

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6.2 Definitions.

6.2.1 Tapping. The phrase "TAPPED LIGHTLY" is defined as meaning the application of a light vibration sufficient to overcome that amount of residual friction normally associated with good quality aircraft instruments.

6.2.2 Hermetic seal. A hermetic seal is defined as a perfectly closed and airtight seal made between vitric or metallic materials, or both. A hermetic seal is not intended to include seals accomplished by gaskets.

6.2.3 "G" unit. The "g" unit is defined as an acceleration of 32.2 feet per second, per second.

6.2.4 Amplitude. Whenever the word "AMPLITUDE" is specified, it shall mean the extent of motion as measured from one extreme to the opposite extreme.

6.3 Ordering data. Procurement documents should specify the following:

a. Whether sampling plan B is to be omitted.

b. Level of packaging and packing desired.

*6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contract or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Oklahoma City ALC/MMEO Tinker AFB, OK 73145 and information pertaining to qualification of products may be obtained from that activity.

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*6.5 Changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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